You must not read the question paper before the exam starts.

You must not write anything until the invigilator says that you may start.

Pay attention to the instructions given by the invigilators.

Read the instructions below **carefully** before the exam starts, **check** you know how you are required to complete your answer booklets or sheets.

You must ensure that the front of your answer books/sheets have been completed correctly **by the end of your exam finish time,** no further time will be given.

January 2026

EC113 2025/6 A 800

Answer Booklet
Student provided non-programmable, non-communication enabled and non-networked calculator permitted

University of Reading

Introductory Microeconomics (EC113)

Three hours

Answer **all** questions from Section A (70 marks) and **one** question from Section B (30 marks).

Formulas/rules provided:

1. Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Chain rule

If $y=f(u(x))$, then $dy/dx=dy/du \times du/dx$

3. Product rule

If $y = u(x)v(x)$, then $\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

4. Quotient rule

If $y = \frac{u(x)}{v(x)}$, then $\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Section A

Answer **all** questions in this section.

1. Use chain rule, product rule *or* quotient rule to differentiate the following functions with respect to x.

   (a) $y = (x^2 + 1)^2$

   (b) $y = \frac{x^2+x-2}{x^3}$

   (10 marks)

2. A monopolist's demand and total cost functions of a good are $P = -Q + 5$ and $C = Q^2$, respectively.

   (a) Derive the total revenue (R) and profit (π) functions.

   (10 marks)

   (b) Calculate the profit maximising output Q* and profit π* using differentiation. (Hint: verify the result with the second-order derivative)

   (10 marks)

3. According to the UK Office for National Statistics (ONS), the price of chocolate rose by 17.7% in the 12 months to May 2025, largely due to reduced cocoa supply from West African producers. Predict the likely changes in the equilibrium price and quantity in the sweets market. Explain your answer. (Hints: demand and supply; show the equilibrium of sweets market before and after the change in chocolate price)

(10 marks)

4. UK food and clothing prices increased by 4.5% and 0.5%, respectively, in the last year. Assume there are only two goods available. Explain how an individual's optimal choice of consumption will change in response to these price changes. (Hint: consumer theory; show the optimal choice before and after the price changes; you can assume no change in the clothing price for simplicity; assume both goods are normal goods and income is unchanged; substitution and income effects)

(10 marks)

5. In a perfectly competitive market, any short-run profit opportunities created by an exogenous increase in demand will disappear in the long run. Explain. (Hints: assume constant-cost industry; start from a long-run equilibrium)

(10 marks)

6. The government increased the National Minimum Wage to £12.21 in 2025 (a 6.7% increase), significantly higher than the Consumer Prices Index (CPI) inflation rate of 3.5% over the same period. How would this policy affect firms' choice of inputs? (Hints: cost minimisation; a higher increase in wages than inflation implies an increase in real wages; assume two inputs, K and L, and the same output level; to simplify the problem, you can focus on firms hiring workers paid at the minimum wage only; show a firm's optimal choice of inputs before and after the change in policy)

(10 marks)

Section B

Answer **one** question only in this section.

7. Given the following demand and supply functions $P = 500 - 2\mathrm{Q}$ and $\mathrm{P} = 200 + Q$, respectively,

   (a) Calculate the equilibrium price and quantity in the market.

   (10 marks)

   (b) Sketch the demand curve, supply curve and your answers to part (a) in one diagram (Hint: for $P \geq 0$ and $Q \geq 0$ only; put P and Q on vertical and horizontal axes, respectively).

   (10 marks)

   (c) Calculate the price elasticities of demand and supply at the market equilibrium using differentiation.

   (10marks)

8. Archie has £100 income. He allocates them among two goods, x and y. The price of x (Px) is 2 and the price of y (Py) is 2. His optimal choice is (25,25).

   (a) Draw the budget constraint and his choice above.

   (10marks)

   (b) 'If x is normal good and Archie's income falls to 80, he must not consume more than 25y.' Explain why this statement is incorrect.

   (20 marks)

(End of Question Paper)